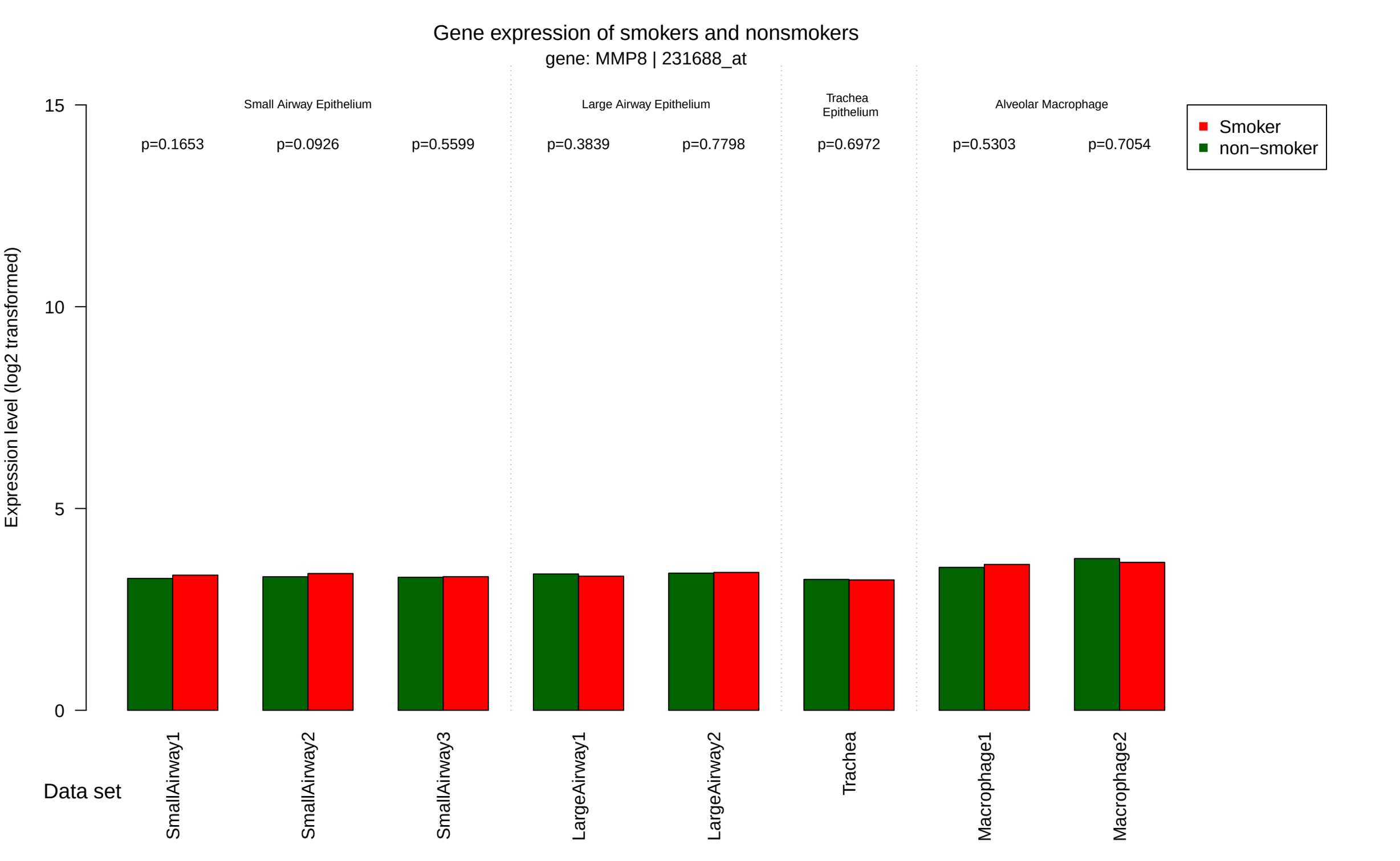
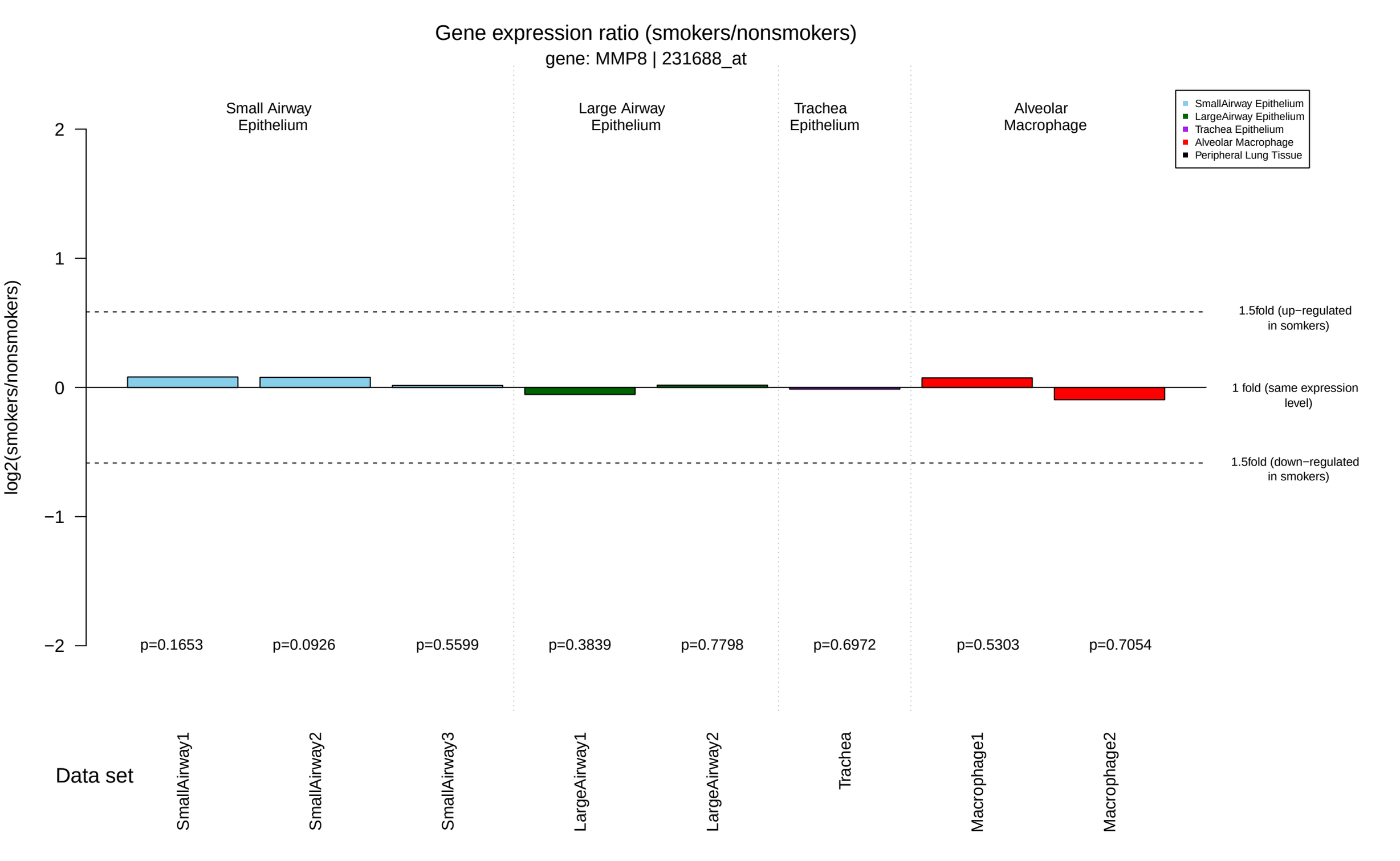
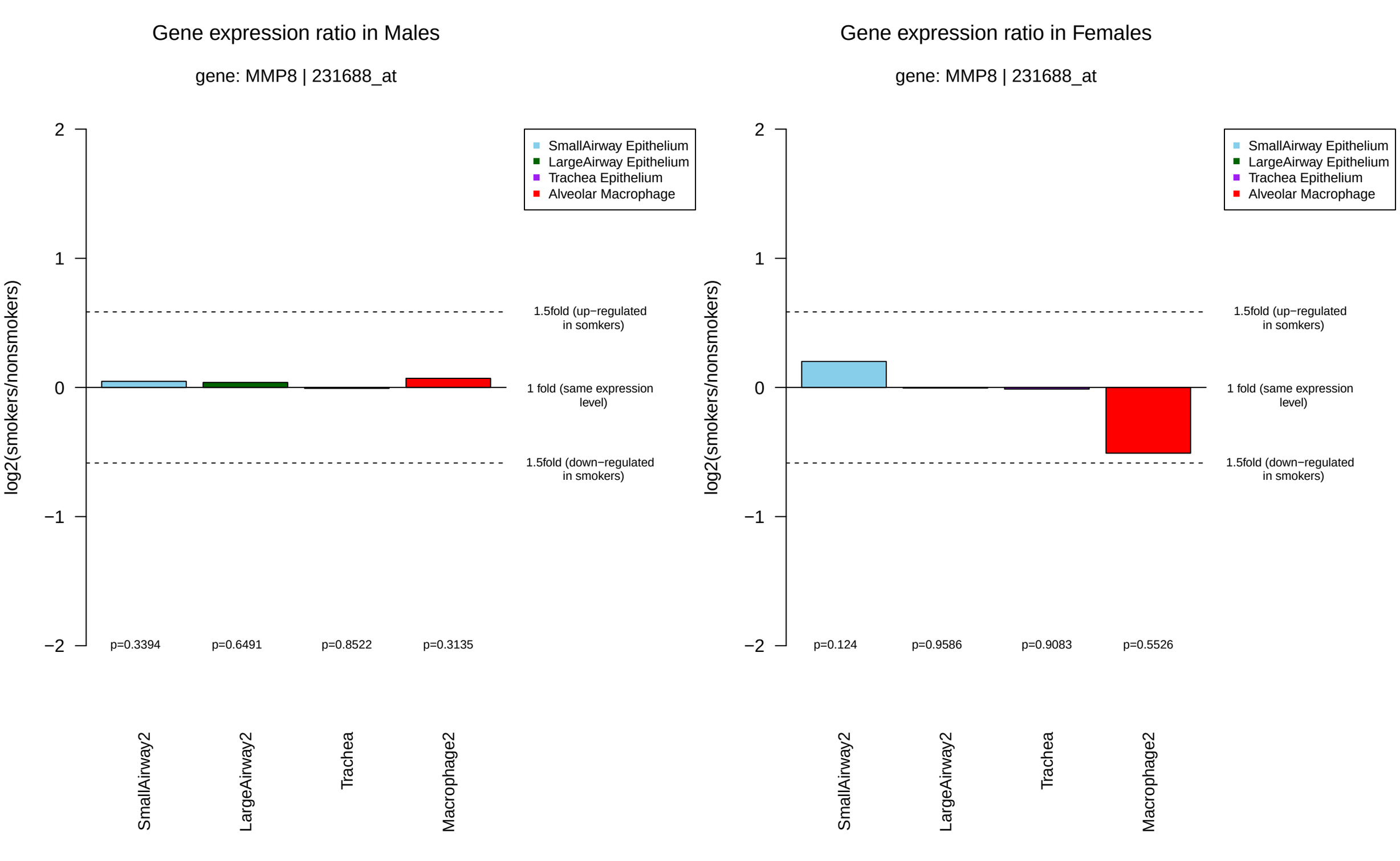
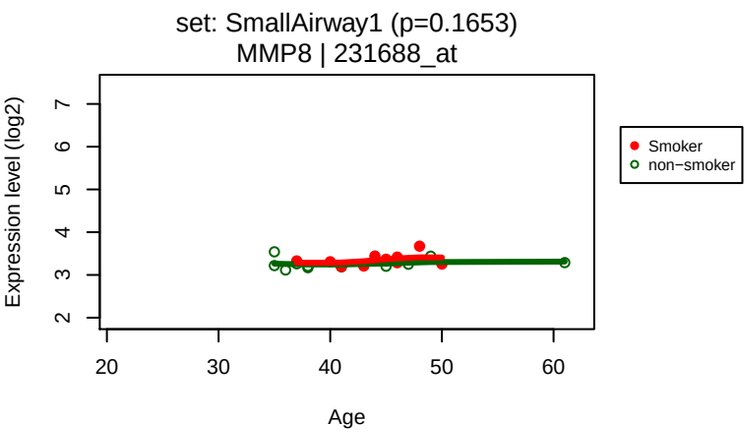
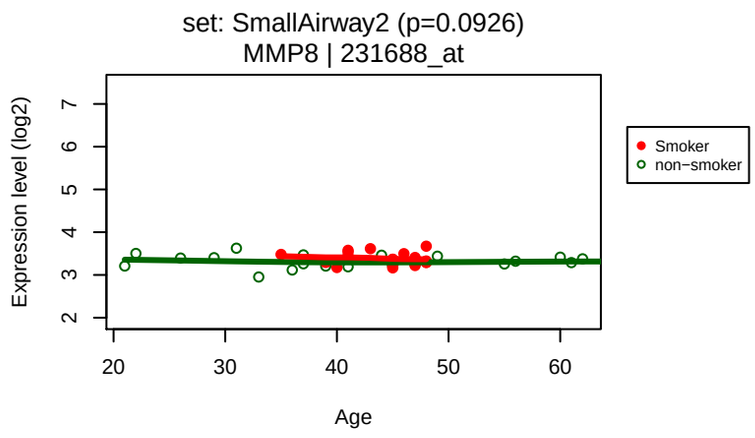
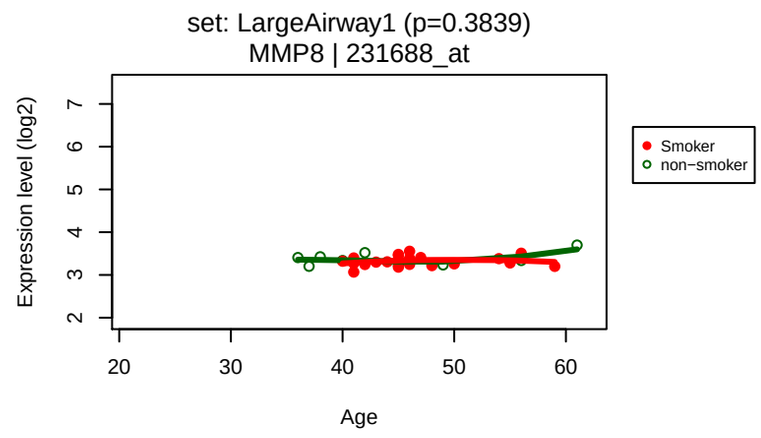

Gene expression of smokers and nonsmokers
gene: MMP8 | 231688_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1653
p=0.0926
p=0.5599
p=0.3839
p=0.7798
p=0.6972
p=0.5303
p=0.7054
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

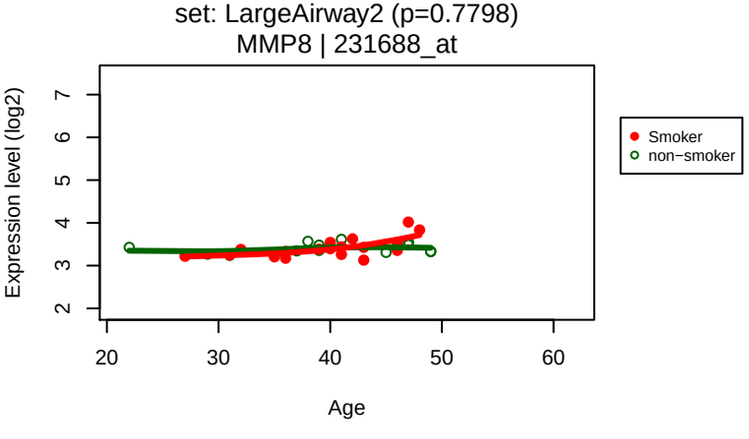

Gene expression ratio (smokers/nonsmokers)
gene: MMP8 | 231688_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1653
p=0.0926
p=0.5599
p=0.3839
p=0.7798
p=0.6972
p=0.5303
p=0.7054
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

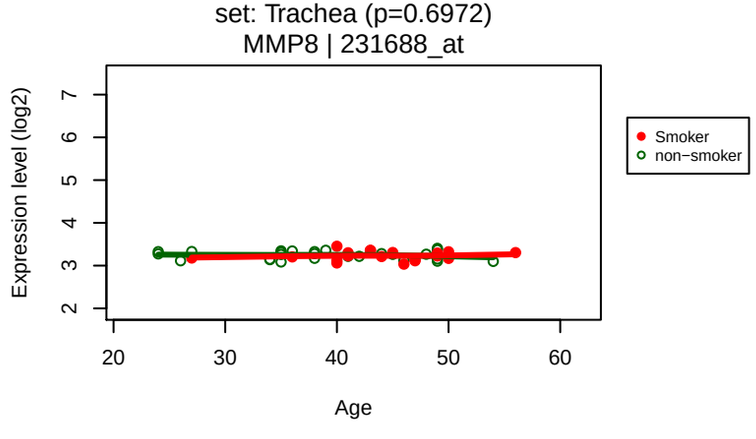

Gene expression ratio in Males
gene: MMP8 | 231688_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3394
p=0.6491
p=0.8522
p=0.3135
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP8 | 231688_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.124
p=0.9586
p=0.9083
p=0.5526
SmallAirway2
LargeAirway2
Trachea
Macrophage2

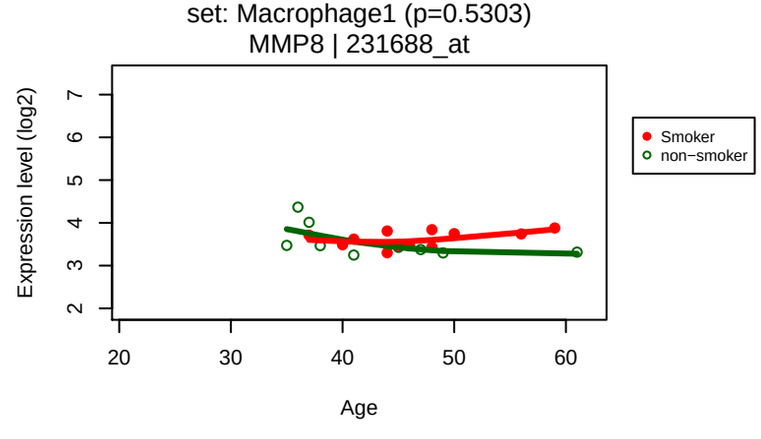

set: SmallAirway1 (p=0.1653)
MMP8 | 231688_at
Expression level (log2)
Age
Smoker
non−smoker

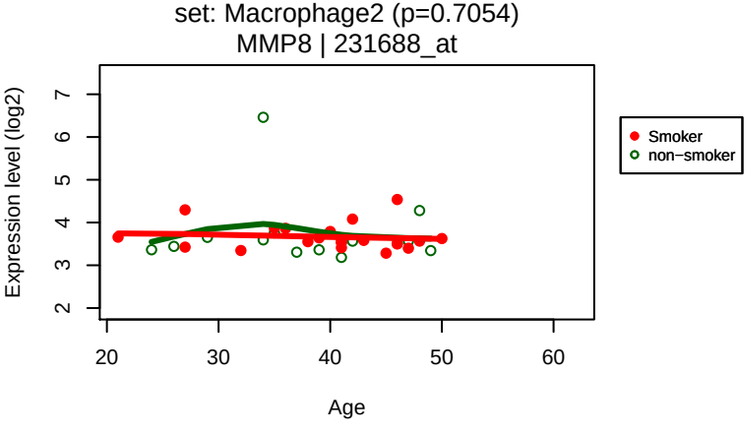

set: SmallAirway2 (p=0.0926)
MMP8 | 231688_at
Expression level (log2)
Age
Smoker
non−smoker

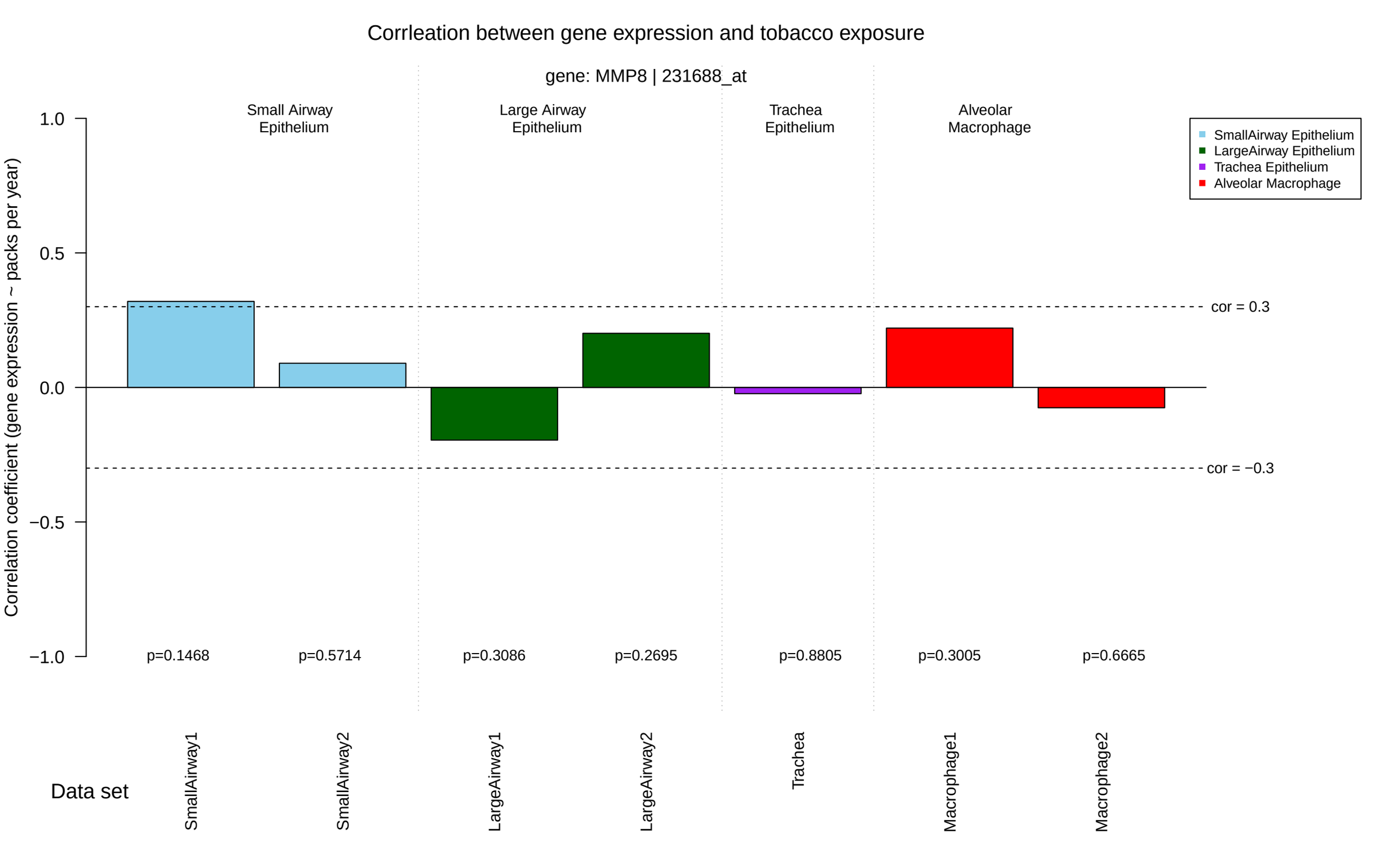

set: LargeAirway1 (p=0.3839)
MMP8 | 231688_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7798)
MMP8 | 231688_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6972)
MMP8 | 231688_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5303)
MMP8 | 231688_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7054)
MMP8 | 231688_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP8 | 231688_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1468
p=0.5714
p=0.3086
p=0.2695
p=0.8805
p=0.3005
p=0.6665
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2